

Capability-Conditioned HetNet

Akshat Guduru

Key References

- HetNet: Learning Efficient Diverse Communication for Cooperative Heterogeneous Teaming
- GATv2: How Attentive are Graph Attention Networks?
- ROMA: Multi-Agent Reinforcement Learning with Emergent Roles
- CASH: Capability-Aware Shared Hypernetworks for Flexible Heterogeneous Multi-Robot Coordination
- HyperMARL: Adaptive Hypernetworks for Multi-Agent RL

HetNet: Overview

- Introduced the Multi-Agent Heterogenous Decentralized Partially Observable Markov Decision Process
- Gives each class its own actor, built from stacked heterogenous Graph Attention (HetGAT) Layers, where every edge carries a short learned binary message.
- A class-wise Multi-Agent Heterogenous Actor Critic trains everything end to end. The critic reads the state summary node which contains global state information, and is present only during training

HetNet: MAH Dec-POMDP

- HetNet generalizes the Dec-POMDP so that different kinds of agent can have different state, observation and action spaces.

$$\left\langle C, \mathcal{N}, \{S^{(i)}\}_{i \in C}, \{A^{(i)}\}_{i \in C}, \{\Omega^{(i)}\}_{i \in C}, \{O^{(i)}\}_{i \in C}, r, T, \gamma \right\rangle$$

Symbol	Meaning
C	Set of agent classes, $i \in C$ names a class
N	Total number of agents, $N^{(i)}$ $i \in C$ number of agents of class i
S	State Vectors of Class i agents
Ω	Class i Observation space
O	Observation Function
r	Reward, shared by all agents
T	Transition Probability Density
γ	Discount

HetNet: Objective

$$\arg \max_{\pi(\bar{s}) \in \Pi} \mathbb{E}_{\pi(\bar{s})} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \middle| \pi(\bar{s}) \right]$$

Choose the per-class policy π^i , one parameter set θ^i per class, that maximized the expected discounted reward.

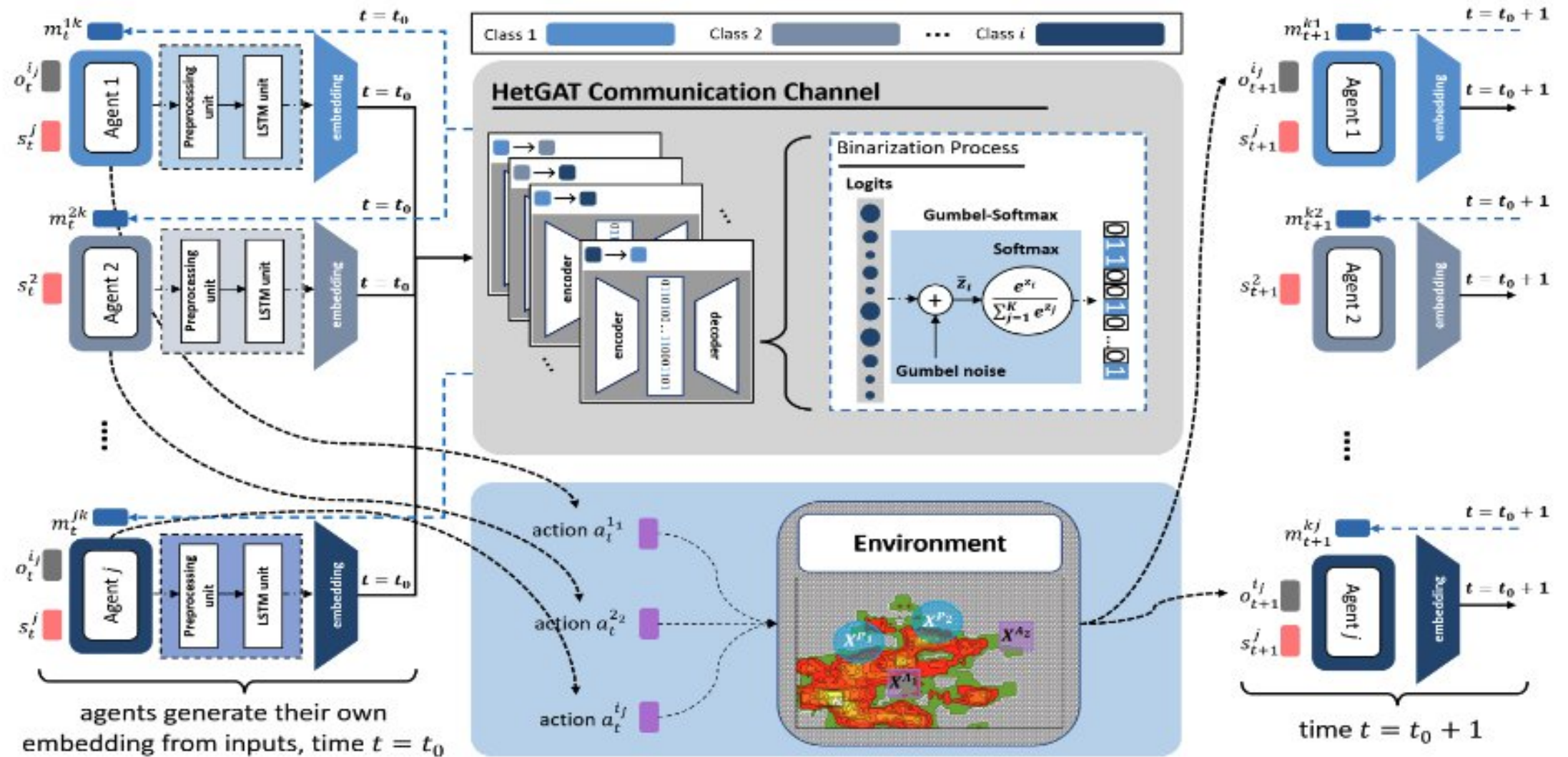


Figure 1: Overview of our multi-agent heterogeneous attentional communication architecture in a CTDE paradigm. At each time point $t = t_0$, each agent j of class i generates a local embedding from its own inputs, by passing its input data through class-specific preprocessing units (i.e., a CNN or a fully-connected NN) and an LSTM cell. Each agent then sends the embedding to a class-specific encoder-decoder networks to generate a binarized message, m_t^{jk} , from its local neighbor k . The message information is decoded and leveraged by the receiving agent to compute the action probabilities as its policy output.

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- For my architecture, I reformulate their question to: Can one shared policy keep HetNet's communication working when the team changes ?
- HetNet requires explicit labels for agent classes, and only shared policies between agents of the same class.
- I ask instead, can we remove the agent classes, and instead introduce agent capabilities.
- Nothing is necessarily new with this architecture, the contribution is a controlled test.

Problem Formulation

- Each agent carries a capability vector.
- Ex. for the PCP env a sensing flag that can fail, and a fixed capture flag:

$$\kappa_{j,t} = (s_{j,t}, a_j) \in \{0,1\}^2$$

- $s_{j,t}$: Agent Sensing Capabilities
- a_j : Agent Capture Capabilities.
- Since action κ is given and not inferred, I mainly aim to study adaptation to known capability changes, not fault diagnosis

